COLLEGE OF ENGENEERING AND TECHNOLOGY

Department:	Computer Engineering		
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Course:	Microprocessor Systems		
Course No.:	CC421	**Sheet #:**	2 Flags and basic assembly

1. Answer the following questions according to the debug screen shown below
   a. What is the address of the next instruction for the following:

AX = 0085 BX = 3300 CX = 0000 DX = 0000 SP = FFEE BP = 0000 SI = 0000 DI = 0000
DS = 0B03 ES = 0B03 SS = 0B03 CS = 0B03 IP = 0106 OV UP EI NG NZ NA PO NC

0B03:0106 7218 JC 0120

   b. If JC is replaced with JO, what is the address of the next instruction? Justify your answers.

Flag	=1	=0
Overflow	OV	NV
Direction	DN	UP
Interrupt	EI	DI
Sign	NG	PL
Zero	ZR	NZA
Auxiliary	AC	NA
Parity	PE	PO
Carry	CY	NC

2. Give the possible range of values of the signed integer N such that when N is added to signed integer 53H, we have no overflow.

Solution: [80H ≤ N ≤ ACH]

3. Give the possible values of 8-bit signed number N that when you add them to the postive number 48H, the sign bit of the result is 1 and the overflow bit is 0

Solution: [80H ≤ N ≤ B7H]

4. Write an assembly snippet to swap the contents of the two registers AX, BX using:
   a. A third register
   b. The Stack
   c. ADD, SUB and NEG
   d. XOR

5. Translate the following pseudo-code into Assembly:

a.
```
If (AX<BX) {
 AX++;
 DX=DX*2;
 CX=CX-AX
}
else {
 BX=CX;
}
```

b.
```
p = 0;
for (i = 15; i > = 0; i--)
{
    p = sum + i;
}
```

c.
```
if (AL > N1) and (BH < N2) and (CL < N3) then
        <statement1>
End if
<statement 2>
```

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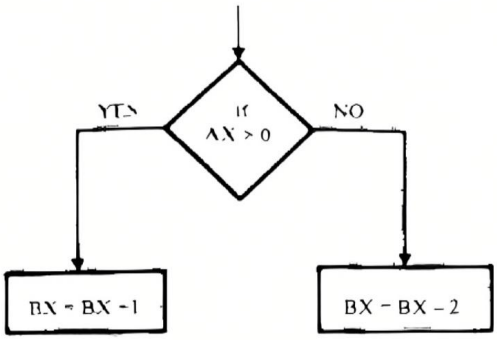
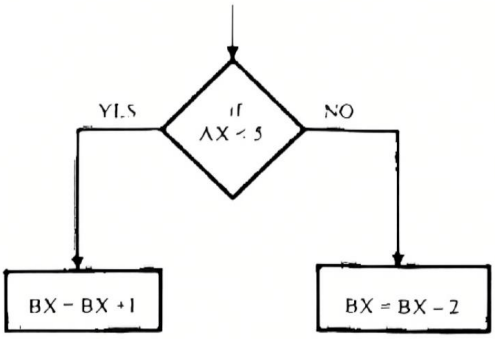
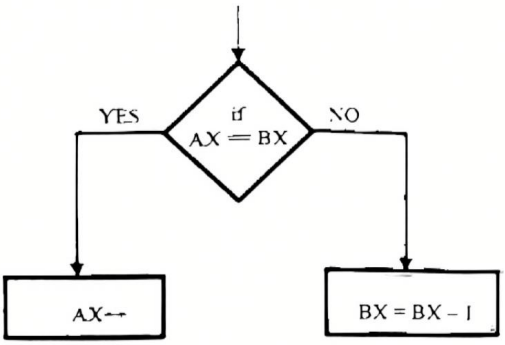
6. Convert to the following flowcharts to Assembly code



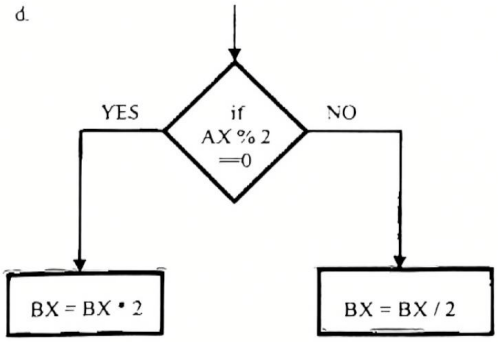


HINT *Check the Sign flag. If "0" then the resultant is greater than "zero", otherwise, it is not.*
Don't forget to update the flag register first, by adding "0" to AX. (You can use the same ***hint*** *in 6 b)*

HINT *Subtract "5" from AX. If the Sign Flag is "1", that means the resultant is lesser than 5, otherwise it is greater*

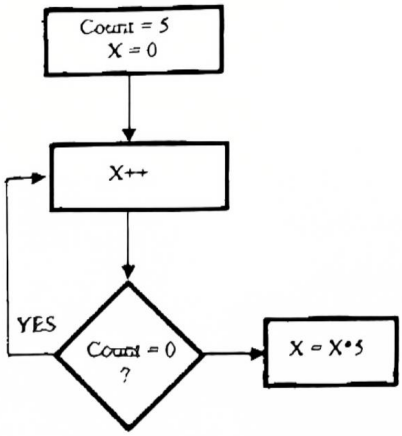


d.



HINT. *Use SUB or XOR. Check if AX == BX by using the Zero Flag*

HINT. *After dividing by 2, check the value in AH, if 0, there is no remainder, otherwise there is. Don't forget to update the Flag Register.*



HINT *Use register CX for the as a counter and any other register to store the value of "X"*

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Basic Assembly Programs

7. Write a complete assembly program to double the contents of the array m, then exchange the contents of array "m" with that of array "n".

m dw 10, 20, 30, 40, 45
n dw 12341H, 45671H, 75311H, 23891H, 87961H

8. Write an assembly program to calculate the sum of five words of data. Each data value represents the yearly wages of a worker. Each worker does not make more than 65535 (FFFFH) a year. The decimal data is as follows 27345, 28521, 29533, 30105 and 32375

9. Write a program that adds the following two multiword numbers and the saves the result.

DATA1 ¬ 548FB9963CE7H
DATA2¬3F65395823411EH

10. Let x defined as follows

x db 05, 19H, 23H, FFH, 12H, 24H

The variable "x" describes an array, which its first element is the length of the array. The other consecutive values are the individual elements of the array. All of these elements are 8-bit signed numbers. Write a program that performs the following mathmatical forumula:

$$\frac{1}{N}\sum_{i=1}^{N} x_i^2$$

where N is the length of the array. Store the result in the location called "result" which size is doubleword. Ignore the fraction part of the calculation.

11. Using Array, write an assembly program that inserts 2500H in position 2. The program must shift all the subsequent numbers two bytes higher in memory to make a room for the inserted number.

Array dw 2345H, 1786H, 5887H, 3421H, 2769H

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